Arman Sakif Chowdhury

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Technical Skills

Programming Languages: Python, SQL, R, Java, JavaScript, C++, C, Dart, Bash.
Data Engineering: ETL/ELT Pipelines, Data Modeling, Change Data Capture (CDC), Data Acquisition, Data Cleaning, PySpark, Apache Spark, Databricks, Microsoft Fabric, SQL Server, Snowflake, Azure Data Lake, SAP Datasphere.
Machine Learning & Data Science: Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Keras, Hugging Face Transformers, NLP, Data Augmentation.
AI & Agentic Systems: MCP Development, LLM Applications, CopilotKit, AG-UI, A2UI, AI Solution Development.
Business Intelligence & Visualization: Power BI, DAX, Power Query, Tableau, Microsoft Excel, Matplotlib, Seaborn, KPI Reporting, Data Storytelling.
Web & Application Development: HTML, CSS, React, Node.js, Django, Flask, Flutter, MongoDB, Selenium, BeautifulSoup, ANTLR.
Cloud, Tools & Practices: Azure, AWS (EC2, ECS, RDS), Kubernetes, Git, CI/CD, Unit Testing, Jupyter Notebook, Google Colab, Postman, Agile/Scrum.

Education

Master of Applied Computing (AI Stream) Sep 2024 – Apr 2026
University of Windsor, Windsor, ON

- Cumulative average: 86.11% — all degree requirements completed (33 credit hours).
- Relevant Coursework: Artificial Intelligence, Natural Language Processing, Advanced Database Topics, Distributed Systems, Advanced Computing Concepts, Advanced Software Engineering.

Bachelor of Science in Computer Science and Engineering Jan 2019 – Jun 2023
Ahsanullah University of Science and Technology, Dhaka, Bangladesh

- CGPA: 3.44 / 4.00 (medium of instruction: English).
- Relevant Coursework: Artificial Intelligence, Pattern Recognition, Soft Computing, Digital Image Processing, Distributed Database Systems, Database Systems, Algorithms, Data Structures, Formal Languages & Compilers, Mathematical Analysis for Computer Science.

Publications

Tackling Fake News in Bengali: Unraveling the Impact of Summarization vs. Augmentation on Pre-trained Language Models 2023

Chowdhury, A. S., Shahariar, G. M., Aziz, A. T., Alam, S. M., Sheikh, M. A., & Belal, T. A. (2023)
arXiv preprint arXiv:2307.06979.

Work Experience

Data Engineer Aug 2026 – Present
FGF Brands, North York, ON

Technologies: Databricks, PySpark, Delta Lake, Unity Catalog, SQL Server, Azure, Python, Power BI

- Build and maintain ETL/ELT pipelines in PySpark on Databricks adopting medallion architecture, including SDP pipeline and CDC logic.
- Model enterprise data from multiple upstream source systems into governed Unity Catalog tables serving downstream analytics and AI workloads.

AI Solutions and BI Developer Co-op Sep 2025 – Aug 2026
FGF Brands, North York, ON
Technologies: Power BI, Databricks, PySpark, SQL Server, Azure, Python, MCP, SAP Datasphere

- Architected a medallion (bronze–silver) lakehouse layer on Databricks Spark Declarative Pipelines with Auto CDC, ingesting Delta Shared Unity Catalog tables from SAP Business Data Cloud through streaming tables and materialized views, replacing direct raw-share consumption and halving end-to-end data latency.
- Built an automated data sync pipeline in PySpark on Databricks to replicate new records from a Databricks view into an on-premise SQL Server table, engineering custom change-data-capture logic that diffs source and destination datasets to append only deltas where no native CDC existed.
- Developed a custom Power BI MCP server on Power BI REST APIs, enabling natural-language querying of enterprise semantic models while enforcing security and answer accuracy through workspace scoping, table-and-column allowlists, and read-only access controls.
- Designed and published interactive Power BI dashboards across Dev, UAT, and Production environments, while redesigning deployment pipelines to reduce deployment time and manual effort by 60%.
- Accelerated pipeline logging from 5+ minutes to near-instant by converting PySpark DataFrames to Pandas after the initial read, eliminating repeated re-materialization of a multi-join upstream view on every logging operation.
- Collaborated across three concurrent project teams in distinct roles: AI adoption researcher on the core AI Solutions team, Power BI developer in the SMI agile pod, and data engineer building master data in SAP Datasphere for a dynamic pricing initiative.

Python Developer Consultant Sep 2023 – Aug 2024
Techscope, New York, USA (Remote)
Technologies: Python, Pandas, SQL, JSON, ANTLR

- Automated the conversion of unstructured client input into structured JSON by authoring custom ANTLR grammars and Python parse-tree visitors, replacing a manual transformation workflow and enabling downstream pipeline ingestion.
- Devised SQL-based data ETL (Extract, Transform, Load) pipelines to optimize data processing workflows on a transactional database of over 100,000 entries.
- Performed comprehensive unit testing using Python's `unittest` framework, reducing error rates by 25%, enabling early bug detection, and enhancing software reliability.
- Engaged with clients directly to gather requirements, analyze business processes, and integrated them into team sprints to facilitate collaboration.

Back-End Developer & Customer Support May 2020 – Aug 2023
ProyogLab, Dhaka, Bangladesh
Technologies: React, Node.Js, Python, Power BI, and Data Warehouse

- Provided hands-on technical support to customers, assisting with Arduino-based hardware projects and troubleshooting embedded systems issues.
- Managed day-to-day sales operations in a startup retail environment, helping scale product distribution from hobbyist kits to educational bundles.
- Played a key role in transitioning company to an online model, contributing to website architecture and backend functionality, including visual dashboard using modern web technologies.
- Developed and maintained backend systems for e-commerce operations, including inventory management, order processing, and data integration. Generated monthly reports using Power BI tools to report progress.

Projects

Predicting Movie Ratings from Plot Summaries [GitHub] Feb 2025 - Mar 2025
Team Research, University of Windsor, ON
Technologies: Python, Pandas, Pytorch, Scikit-learn, Selenium and BeautifulSoup

- Conceptualized a Large Language Model (BERT) to predict movie success using pre-release data, specifically only using plot summaries as input.

- Utilized two publicly available corpora and created a custom dataset with 17,877 records by scraping the IMDB website, ensuring a comprehensive dataset for analysis.
- Exceeded baseline model performance by 11% with our 81% F1-score and deployed the optimized model on Huggingface Spaces for seamless accessibility and evaluation.

Mitigating Bias in Natural Language Understanding using GNN [GitHub] Sep 2024 - Dec 2024
Team Research, University of Windsor, ON

Technologies: Python, Pandas, Pytorch, Scikit-learn, and Kagglehub

- Formulated a Graph Neural Networks (GNN) model to mitigate racial bias in natural language understanding by either removing less correlated sensitive attributes or assigning lower weights to them.
- Demonstrated the feasibility of building a computationally expensive model on low-resource platforms by utilizing only a CPU on Google Colaboratory.
- Maintained competitive performance by achieving 87.47% accuracy, closely matching state-of-the-art models (91%) that do not employ data anonymization.

Bengali Fake News Detection [Journal] Aug 2022 - Sep 2025
Team Research, Ahsanullah University of Science and Technology, Bangladesh

Technologies: Python, PyTorch, Hugging Face Transformers, BERT, NLP, Data Augmentation

- Led a four-member research team through peer review to a first-author journal publication, classifying fake news in low-resource Bengali using five pre-trained transformer language models.
- Fine-tuned BanglaBERT and mBERT with summarization and data augmentation pipelines, achieving 96–97% accuracy on two benchmark test sets and 86% on a third set reserved for generalization evaluation.
- Curated an expanded training corpus by translating English news articles to offset Bengali fake-news scarcity, and summarized long articles to overcome BERT's token-length limitations.

Churn BigML: Telecom Customer Churn Prediction [GitHub] Feb 2023 - May 2023
Team Research, University of Windsor, ON

Technologies: Python, Scikit-learn, LightGBM, XGBoost, Pandas

- Benchmarked eight classification algorithms (KNN, SVM, Random Forest, AdaBoost, Gradient Boosting, LGBM, XGBoost) on a 3,333-instance telecom dataset, reaching 95.74% accuracy with LGBM and XGBoost.
- Identified the strongest churn drivers through feature importance and correlation analysis, translating model results into actionable customer retention insights.

Technical Training

Agile Foundations – The Rise of Knowledge Workers (LinkedIn Learning) Jan 2025

- Acquired hands-on understanding of Agile principles, Scrum roles, and iterative development, emphasizing team collaboration, adaptability, and continuous improvement in dynamic, knowledge-driven environments.

Flutter and Dart – The Complete Guide Oct 2024
Academind by Maximilian Schwarzmüller (Udemy)

Ultimate AWS Certified Solutions Architect (Udemy) Jun 2024

- Gained experience in designing and deploying scalable, secure, and cost-effective AWS cloud solutions and became familiar with services slike EC2, S3, Lambda, RDS and IAM.

Spark and Python for Big Data with PySpark (Udemy) Oct 2023

- Enhanced proficiency in distributed computing, data wrangling, optimization techniques for big data workflows, and overall knowledge on processing and analyzing large-scale datasets

100 Days of Code: The Complete Python Pro Bootcamp (Udemy) Sep 2023

- Strengthened Python programming skills through hands-on projects covering automation, web development, and data science while using libraries like Flask, Pandas, Selenium, and TensorFlow for real-world applications.

Extracurricular Activities

VentureU Bootcamp Feb 18-22, 2025
Office of Innovation, Partnerships and Entrepreneurship, University of Windsor, ON

- Benefited from expert-led 5-day-long bootcamp, enhancing problem-solving capabilities and fostering the development of practical, applicable solutions.
- Participated in team-based problem-solving and demonstrated presentation skills through competitive pitch competitions, demonstrating the ability to effectively communicate complex ideas.

Event Volunteer Oct 2024 - Present
University Community Church, Windsor, ON

- Assisted with setup, serving, dishwashing and cleanup of Sunday meals for students.

International Students Mentor Dec 2024 - Jan 2025
International Student Centre, University of Windsor, ON

- Mentored 15 international students at the University of Windsor by answering questions and assisting with transition.